Netskope Secure Web Gateway (SWG) Deployment & Web Filtering

Implementation Solution Guide

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Contents

Executive Summary — Netskope Secure Web Gateway (SWG) Deployment & Web Filtering 4

What you get . . . 4

What this delivers . . . 4

The deployment model . . . 4

Outcome . . . 4

1. Solution Overview — Netskope Secure Web Gateway (SWG) 5

What a Secure Web Gateway is . . . 5

How it works . . . 5

What you can control . . . 5

The named artifacts this solution creates . . . 5

Benefits . . . 6

Limitations (be honest with buyers) . . . 6

Why pilot-first . . . 6

The standard deployment model . . . 7

2. Requirements Guide — Netskope Secure Web Gateway (SWG) 7

2.1 Netskope SWG module licensed . . . 7

2.2 Identity-based steering ready . . . 7

2.3 Steering-scope groups . . . 7

2.4 Client deployment to managed devices . . . 7

2.5 Decryption (root) certificate trusted on devices . . . 8

2.6 Network and egress allow the gateway . . . 8

2.7 SSL decryption bypass list prepared . . . 8

2.8 Change control and rollback (inline traffic) . . . 8

Checklist . . . 8

3. Implementation Guide — Netskope Secure Web Gateway (SWG) 9

Step 1 — Confirm tenant and identity . . . 9

Step 2 — Create the steering-scope groups (Entra) . . . 9

Step 3 — Deploy the decryption certificate (Intune) . . . 9

Step 4 — Deploy the Netskope client (Intune) . . . 10

Step 5 — Build the URL categorization profile (Netskope) . . . 10

Step 6 — Configure SSL decryption with the bypass list (Netskope) . . . 10

Step 7 — Create the threat and content control (Netskope) . . . 10

Step 8 — Create the web-filtering policy (Netskope) . . . 11

Step 9 — Steer the pilot scope . . . 11

Step 10 — Validate the pilot (gate) . . . 11

Step 11 — Expand the steering scope . . . 11
5. Testing Guide — Netskope Secure Web Gateway (SWG) 12
Test 1 — Client deployed and steering active . . . 12
Test 2 — Decryption certificate trusted . . . 12
Test 3 — Categorization and block action . . . 12
Test 4 — Coach (warn) action . . . 12
Test 5 — Allowed traffic flows normally . . . 12
Test 6 — Decryption bypass honored . . . 12
Test 7 — On/off-network consistency . . . 12
Test 8 — Coverage and activity report . . . 13
Pass/fail gate . . . 13
6. Troubleshooting Guide — Netskope Secure Web Gateway (SWG) 13
A. Device not steered / traffic bypassing the gateway . . . 13
B. HTTPS certificate errors . . . 14
C. An app breaks under inspection . . . 14
D. Wrong block / site miscategorized . . . 14
E. Coaching/notification page not shown . . . 14
F. Off-network policy inconsistent with on-network . . . 14
G. Diagnostics quick reference (device side) . . . 14
H. When to escalate . . . 14
7. Best Practices — Netskope Secure Web Gateway (SWG) 15
8. Deliverables Summary — Netskope Secure Web Gateway (SWG) 15
Documentation . . . 16
PowerShell scripts . . . 16
What the buyer receives . . . 16
Outcome . . . 16
Appendix A — PowerShell Scripts 18

Executive Summary — Netskope Secure Web Gateway (SWG) Deployment & Web Filtering

Web traffic is where most threats and most data loss start, yet many organizations still rely on patchy filtering that remote users bypass the moment they leave the office. The result is inconsistent policy, threats that slip in over HTTPS, and no control once a laptop is off the corporate network.

The `Netskope Secure Web Gateway (SWG)` solves this by putting every user's web traffic behind one cloud-delivered chokepoint: URL categorization, a web-filtering policy, threat and content controls, and consistent enforcement whether the user is on or off the corporate network. This package stands that gateway up the repeatable way — pilot-first and rollback-ready, because inline proxy affects live traffic.

This is a complete, deploy-it-today package written for a competent network/security engineer with no prior Netskope experience. The Microsoft side (identity, client and certificate delivery, and steering scope) is given as exact navigation; the Netskope console internals are given as the correct architectural and procedural steps with the precise control named, and deferred to Netskope's official documentation for the exact current screen.

What you get

- A plain-language `Solution Overview` of how a cloud SWG works: traffic steering, URL categorization, web-filtering policy, threat/content controls, and on/off-network enforcement.
- A `Requirements Guide` with verify / implement-if-missing for every prerequisite.
- An `Implementation Guide` of precise, ordered steps across Netskope, Entra ID, and Intune.
- Five `PowerShell scripts` for readiness discovery, client/steering validation, a web-traffic and policy-coverage report, safe remediation of mis-steered devices, and a steering-scope group builder.
- `Testing` and `Troubleshooting` guides with explicit pass/fail gates.

What this delivers

A cloud secure web gateway with a named, reusable set of artifacts — a steering scope, a URL categorization profile, a web-filtering policy, and threat/content controls — proven on a pilot group before it is expanded fleet-wide, with a documented rollback at every stage.

The deployment model

`Assess` → `Configure` → `Pilot` → `Validate` → `Expand`, with a pilot gate before any fleet rollout.

Outcome

Every user's web traffic behind one cloud secure web gateway — categorized, filtered, and consistently enforced wherever they work — proven on a pilot before you scale.

1. Solution Overview — Netskope Secure Web Gateway (SWG)

What a Secure Web Gateway is

A Secure Web Gateway (SWG) is a cloud-delivered inline proxy that sits between your users and the web. Every web request is steered to the gateway, where it is identified (which user/group), decrypted (for HTTPS inspection), categorized (what kind of site), and evaluated against a web-filtering policy before it is allowed, blocked, or coached. The Netskope SWG provides this as part of a SASE/SSE platform, so the same policy follows the user on or off the corporate network.

How it works

1. Traffic is **steered** into the Netskope cloud. On managed devices this is done by the Netskope client (steering config from your tenant); on networks it can be done by a forward proxy or tunnel. This solution uses **client-based steering** delivered to managed devices.

1. The gateway **identifies the user** via the identity tied to the client enrollment / SSO, so policy can be written per user and per group.

1. For HTTPS sites, the gateway performs **SSL/TLS decryption** (inspection) so it can categorize and filter encrypted traffic. A decryption (root) certificate must be trusted on the device.

1. The request URL is matched to a **URL category** (for example, malware, phishing, gambling, social media, productivity) using the gateway's categorization engine.

1. A **web-filtering policy** evaluates the category, the user/group, and threat/content signals, then **allows, blocks, or coaches** (warns the user) the request.

1. Activity is logged in the tenant, giving you per-user, per-category web visibility and the data the reporting script summarizes.

What you can control

• **URL categorization** — which categories exist and how sites are classified (a categorization profile that the filtering policy references).

• **Web-filtering policy** — allow / block / coach by category, user, and group; the order of rules.

• **Threat and content controls** — block known-malicious categories, risky file types, and threat-classified destinations.

• **SSL/TLS decryption scope** — which traffic is inspected vs. bypassed (for privacy/compatibility).

• **Steering scope** — which users/devices are steered into the gateway, staged from pilot to fleet.

The named artifacts this solution creates

Use these names verbatim across the console, the scripts, and your change records:

Artifact	Name	Where
Entra steering scope (pilot)	`Web Steering – Pilot – Stage 1`	Entra group
Entra steering scope (broad)	`Web Steering – Broad – Stage 2`	Entra group
Client deployment (app)	`SWG Client Deployment – Managed Devices – Production`	Intune app
Decryption certificate profile	`Decryption Certificate – Managed Devices – Production`	Intune cert profile
URL categorization profile	`URL Categorization Profile – Standard – Production`	Netskope
Web-filtering policy	`Web Filtering Policy – All Users – Production`	Netskope
Threat/content control	`Threat & Content Control – All Users – Production`	Netskope

Benefits

- **One chokepoint** — every user's web traffic is categorized and filtered in one place.
- **Consistent on/off network** — the same policy applies wherever the user works.
- **Threat reduction** — known-malicious categories and risky content are blocked inline.
- **Visibility** — per-user, per-category web activity for reporting and investigation.
- **Repeatable** — assign the steering scope once, expand everywhere, with a pilot gate.

Limitations (be honest with buyers)

- An SWG is an **inline proxy**: it sits in the live traffic path, so changes affect users in real time and must be piloted and reversible.
- **HTTPS inspection requires a trusted decryption certificate** on the device; sites with certificate pinning may need decryption bypass.
- Categorization is **best-effort**; miscategorized sites need exception handling.
- The SWG controls **web (and, depending on licensing, cloud-app) traffic** — it is not a full firewall replacement for all protocols.

Why pilot-first

Because the gateway is inline, a wrong block or a decryption-compatibility problem hits real users immediately. Steering a small `Web Steering – Pilot – Stage 1` group first lets you validate enforcement, user experience, and the rollback before any of the fleet is affected.

The standard deployment model

> ***Confirm tenant + identity → configure categorization, filtering, and threat/content policy → steer the pilot scope → validate enforcement and user experience → expand the steering scope fleet-wide.*** *Model:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Netskope Secure Web Gateway (SWG)

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-network.md` (a Netskope tenant, Entra ID, Intune, admin roles, network/identity fundamentals, a pilot group, and change control with a rollback path).

2.1 Netskope SWG module licensed

Description: The Netskope tenant has the Secure Web Gateway module (and SSL inspection) licensed. **Why:** Web filtering, categorization, and threat controls are SWG features. **Verify:** In the Netskope console, confirm the SWG / web policy and SSL decryption controls are available to your admin; or run `NSSWG-Discovery.ps1` (queries tenant readiness via the REST API). **Implement:** Engage your Netskope account team to enable the SWG and SSL inspection modules.

2.2 Identity-based steering ready

Description: Users/groups are present in Entra ID and the gateway can resolve user identity. **Why:** Per-user/per-group web-filtering policy requires identified traffic. **Verify:** Entra → **Users** / **Groups** show the target population; the Netskope tenant is integrated with your identity provider (SSO / directory). **Implement:** Sync directory to Entra ID; configure the Netskope tenant's identity integration per Netskope's documentation (the exact integration screen is deferred to the vendor).

2.3 Steering-scope groups

Description: Entra security groups `Web Steering – Pilot – Stage 1` and `Web Steering – Broad – Stage 2`. **Why:** These scope who is steered into the gateway, staged pilot then broad. **Verify:** Entra → **Groups**; both groups exist; or run `NSSWG-Validation.ps1`. **Implement:** Create them with `NSSWG-Set-ScopeTargeting.ps1`, or Entra → **Groups** → **New group** → **Security**.

2.4 Client deployment to managed devices

Description: The Netskope client (steering agent) deployed to target devices via Intune as `SWG Client Deployment – Managed Devices – Production`. **Why:** The client steers device web traffic into the gateway and applies the steering config. **Verify:** Intune → **Apps** → **Windows**; the app exists and shows installs; or run `NSSWG-Validation.ps1` (checks the client/service on the device). **Implement:** Package

the Netskope client per Netskope's deployment guidance and deploy it from Intune (the client package and its tenant parameters are obtained from your Netskope tenant).

2.5 Decryption (root) certificate trusted on devices

Description: The Netskope SSL decryption certificate deployed via Intune as `Decryption Certificate – Managed Devices – Production` and trusted in the device root store. **Why:** HTTPS inspection fails (certificate errors) unless the decryption certificate is trusted. **Verify:** On a device, the certificate is in **Trusted Root Certification Authorities**; or run `NSSWG-Validation.ps1`. **Implement:** Export the certificate from your Netskope tenant and deploy it as a Trusted Certificate profile in Intune.

2.6 Network and egress allow the gateway

Description: Outbound HTTPS to the Netskope cloud is permitted; no upstream proxy strips/blocks it. **Why:** Steered traffic must reach the gateway; an upstream proxy can break steering or double-proxy. **Verify:** From a target device, the Netskope client connects (steering active); DNS resolves and the internet is reachable. **Implement:** Allow the Netskope cloud endpoints through firewall/proxy per Netskope's published ranges; remove conflicting upstream proxy settings for steered devices.

2.7 SSL decryption bypass list prepared

Description: A list of sites/categories to **exclude** from decryption (certificate-pinned apps, sensitive/regulated categories such as health or banking, per your policy). **Why:** Some apps break under inspection, and some categories should not be decrypted for privacy. **Why it matters at planning time:** decrypting everything on day one causes breakage and complaints. **Verify:** A documented bypass list is approved before enabling decryption broadly. **Implement:** Build the bypass list and apply it in the decryption policy (the exact control is named in the Implementation Guide; the screen is deferred to Netskope's documentation).

2.8 Change control and rollback (inline traffic)

Description: A change window and a written rollback — remove the pilot scope from steering, disable the filtering policy, or uninstall the client. **Why:** Inline proxy and decryption affect live traffic; every stage must be reversible. **Verify:** The rollback is written and approved before steering any production users (see G7). **Implement:** Document the rollback and a user-comms plan; rehearse removing `Web Steering – Pilot – Stage 1` from steering.

Checklist

- Netskope SWG + SSL inspection licensed — `NSSWG-Discovery.ps1`
- Identity integration ready; users/groups in Entra ID
- Steering groups `Web Steering – Pilot – Stage 1` / `Web Steering – Broad – Stage 2`
- Client `SWG Client Deployment – Managed Devices – Production` deployed
- Certificate `Decryption Certificate – Managed Devices – Production` trusted

- Network/egress allows the Netskope cloud
- SSL decryption bypass list prepared and approved
- Change window + rollback approved

3. Implementation Guide — Netskope Secure Web Gateway (SWG)

Follow these steps **in order**. Microsoft-side steps give exact navigation in the Microsoft Entra admin center (`https://entra.microsoft.com`) and the Microsoft Intune admin center (`https://intune.microsoft.com`). Netskope-side steps name the exact control to configure but defer the precise current console screen to Netskope's official documentation — find the named control in your tenant.

> ***Model:*** *Configure categorization, filtering, and threat/content policy first; steer only the* `Web Steering – Pilot – Stage 1` *scope; validate enforcement and user experience; then expand the steering scope to* `Web Steering – Broad – Stage 2`*. The gateway is inline — never steer production users before the pilot gate passes.* ***Rollback at any stage:*** *remove the affected scope group from steering, or disable the* `Web Filtering Policy – All Users – Production` *policy, or uninstall the client — restoring direct-to-internet traffic.*

Step 1 — Confirm tenant and identity

1. Sign in to the Netskope console as an administrator and confirm the SWG and SSL inspection modules

are available (the named area is the web/SWG policy and the SSL decryption policy).

1. Confirm the tenant's identity integration resolves users (SSO / directory). The exact integration

screen is deferred to Netskope's documentation; verify by confirming a test user is identified.

1. Run `NSSWG-Discovery.ps1` to record tenant readiness before changing anything.

Step 2 — Create the steering-scope groups (Entra)

1. Entra → **Groups** → **New group** → **Group type: Security** → **Name:** `Web Steering – Pilot – Stage 1`

→ add a few representative pilot devices/users → **Create**.

1. Repeat for `Web Steering – Broad – Stage 2` (the rest of the fleet) → **Create**.

(You can create both with `NSSWG-Set-ScopeTargeting.ps1`.)

Step 3 — Deploy the decryption certificate (Intune)

1. Export the Netskope SSL decryption (root) certificate from your tenant (the certificate-download

control; screen deferred to Netskope's documentation).

1. Intune → **Devices** → **Configuration** → **Create** → **New policy** → **Platform: Windows 10 and later** →

Profile type: Templates → **Trusted certificate** → **Create**.

1. **Name:** `Decryption Certificate – Managed Devices – Production`. Upload the certificate;

Destination store: Computer certificate store – Root. **Next**.

1. **Assignments** → **Add groups** → `Web Steering – Pilot – Stage 1`. **Next** → **Create**.

Step 4 — Deploy the Netskope client (Intune)

1. Obtain the Netskope client package and its tenant enrollment parameters from your Netskope tenant (the client-download/parameters control; screen deferred to Netskope's documentation).

1. Intune → **Apps** → **Windows** → **Add** → choose the appropriate Windows app type for the package →

upload the package and set the install/enrollment parameters.

1. **Name:** `SWG Client Deployment – Managed Devices – Production`.
2. **Assignments** → **Required** → **Add groups** → `Web Steering – Pilot – Stage 1`. **Create.**

Step 5 — Build the URL categorization profile (Netskope)

1. In the Netskope console, create a URL categorization profile named `URL Categorization Profile – Standard – Production`.

1. Select the categories your `Web Filtering Policy` will reference — at minimum a block set (malware, phishing, command-and-control, and other security categories) and any acceptable-use categories (for example gambling, adult, illegal) per your policy.

1. The exact categorization screen is deferred to Netskope's documentation; the control to find is the URL category list / custom categories used by web policy.

Step 6 — Configure SSL decryption with the bypass list (Netskope)

1. In the SSL decryption policy, enable decryption for steered traffic so HTTPS can be inspected.
2. Apply the **decryption bypass list** prepared in Requirement 2.7 (certificate-pinned apps and sensitive/regulated categories) so those flows are **not** decrypted.

1. The control to find is the SSL decryption (inspection) policy and its bypass/exception list; the screen is deferred to Netskope's documentation.

Step 7 — Create the threat and content control (Netskope)

1. Create a threat/content control named `Threat & Content Control – All Users – Production` that

blocks threat-classified destinations and risky file types for steered traffic.

1. The control to find is the threat-protection / file-type policy applied to web traffic; the screen

is deferred to Netskope's documentation.

Step 8 — Create the web-filtering policy (Netskope)

1. Create a web-filtering policy named `Web Filtering Policy – All Users – Production` that references

`URL Categorization Profile – Standard – Production`.

1. Define rule actions: **block** the security/unacceptable categories, **coach** (warn) any

borderline categories per your policy, and **allow** the rest.

1. Scope the policy to the steered users (identity-based). Order the rules so block rules precede

allow rules. The control to find is the real-time/web protection policy; the screen is deferred to Netskope's documentation.

Step 9 — Steer the pilot scope

1. In the Netskope steering configuration, add `Web Steering – Pilot – Stage 1` to the active steering

scope so only pilot devices are steered into the gateway. The control to find is the steering configuration / user-group scope; the screen is deferred to Netskope's documentation.

1. On a pilot device, confirm the client shows steering active and the decryption certificate is

trusted; run `NSSWG-Validation.ps1`.

Step 10 — Validate the pilot (gate)

1. Run `NSSWG-Validation.ps1` on pilot devices → client running, steering active, certificate trusted.
2. Run `NSSWG-Report.ps1` → confirms pilot web activity is flowing through the gateway and policy

coverage (which steering groups exist and are scoped).

1. Browse known-category test sites from a pilot device and confirm block/coach/allow behavior matches

`Web Filtering Policy – All Users – Production`. Complete the Testing Guide.

1. **Pilot gate:** do not proceed until the pilot passes and no user-impacting breakage remains.

Step 11 — Expand the steering scope

1. Add `Web Steering – Broad – Stage 2` to the active steering scope in Netskope.
2. Add `Web Steering – Broad – Stage 2` to the **Assignments** of

`SWG Client Deployment – Managed Devices – Production` and `Decryption Certificate – Managed Devices – Production` in Intune so broad devices receive the client and certificate.

1. Re-validate with `NSSWG-Report.ps1`; remediate any mis-steered or non-compliant devices with `NSSWG-Remediation.ps1` (default `-WhatIf`).

Order of operations

`Confirm tenant + identity → steering groups → certificate → client → categorization profile → SSL decryption + bypass → threat/content control → web-filtering policy → steer pilot → validate (gate) → expand`

5. Testing Guide — Netskope Secure Web Gateway (SWG)

Run after the pilot scope is steered (Implementation Step 10) and again after expanding to broad.

Test 1 — Client deployed and steering active

Do: On a pilot device run `NSSWG-Validation.ps1` (checks the client/service and steering state). **Expected:** The Netskope client is installed and running and steering is active.

Test 2 — Decryption certificate trusted

Do: `NSSWG-Validation.ps1`, or check **Trusted Root Certification Authorities** on the device. **Expected:** `Decryption Certificate – Managed Devices – Production` is present and trusted; HTTPS sites load without certificate errors through the gateway.

Test 3 — Categorization and block action

Do: From a pilot device, browse a site in a category you set to **block** in `Web Filtering Policy – All Users – Production` (use a known test category, not a live malicious site). **Expected:** The request is blocked and the gateway block/notification page is shown.

Test 4 — Coach (warn) action

Do: Browse a site in a category set to **coach**. **Expected:** The user sees the coaching/warning page and can proceed per policy.

Test 5 — Allowed traffic flows normally

Do: Browse normal business sites in allowed categories. **Expected:** Sites load normally; the activity appears in the tenant's web logs.

Test 6 — Decryption bypass honored

Do: Open an app/site on the decryption bypass list (certificate-pinned or sensitive category). **Expected:** The flow works without inspection errors (not decrypted).

Test 7 — On/off-network consistency

Do: Repeat Tests 3 and 5 with the device off the corporate network (e.g. on a home/hotspot link). **Expected:** The same block/allow behavior applies off-network as on-network.

Test 8 — Coverage and activity report

Do: Run `NSSWG-Report.ps1`. **Expected:** Steered scope groups are listed and pilot web activity (per-category counts) is present, confirming traffic is flowing through the gateway.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–7 pass on `Web Steering – Pilot – Stage 1`; no user-impacting breakage; rollback rehearsed
Broad pass	Tests 1–3, 5, 8 pass for `Web Steering – Broad – Stage 2` after expansion

6. Troubleshooting Guide — Netskope Secure Web Gateway (SWG)

Use `NSSWG-Validation.ps1` for device state, `NSSWG-Discovery.ps1` for readiness, `NSSWG-Report.ps1` for coverage/activity, and `NSSWG-Remediation.ps1` (default `-WhatIf`) for safe fixes. The exact Netskope console screens referenced below are deferred to Netskope's documentation; the named control is given so you can find it in your tenant.

A. Device not steered / traffic bypassing the gateway

Symptom	Cause	Remediation
Web activity not appearing in the tenant for a device	Client not installed, or device not in a steered scope	Run `NSSWG-Validation.ps1`; if not steered, add the device to `Web Steering – Pilot – Stage 1` / `Web Steering – Broad – Stage 2` and re-sync with `NSSWG-Remediation.ps1`
Client installed but steering inactive	Client not enrolled to the tenant, or upstream proxy interfering	Re-deploy `SWG Client Deployment – Managed Devices – Production`; remove conflicting upstream proxy for steered devices
Steering active on-network only	Steering scope/config not applied off-network	Confirm the steering configuration covers off-network in the Netskope steering control

B. HTTPS certificate errors

- Browsers show certificate warnings on inspected sites → the decryption certificate is not trusted.

Confirm `Decryption Certificate – Managed Devices – Production` is in **Trusted Root Certification Authorities** (run `NSSWG-Validation.ps1`); re-deploy the Intune trusted-certificate profile.

C. An app breaks under inspection

- A certificate-pinned app fails only when steered/decrypted → add it to the **SSL decryption bypass list** (the SSL decryption policy's exception control) so the flow is not inspected. Re-test.

D. Wrong block / site miscategorized

Symptom	Cause	Remediation
A legitimate site is blocked	Category set to block, or site miscategorized	Confirm the category in `URL Categorization Profile – Standard – Production`; submit a category-correction request and add a documented allow exception in `Web Filtering Policy – All Users – Production`
A site that should be blocked is allowed	Rule order or missing category	Reorder so block rules precede allow rules; add the category to the block set

E. Coaching/notification page not shown

- A coached category just blocks or allows with no page → confirm the rule action is **coach** (not block/allow) and that the gateway notification template is enabled for that action.

F. Off-network policy inconsistent with on-network

- Off-network users get different behavior → confirm the same `Web Filtering Policy – All Users – Production` and steering scope apply regardless of network; identity-based (not IP-based) policy follows the user.

G. Diagnostics quick reference (device side)

```
Get-Service | Where-Object { $_.Name -like '*stagent*' -or $_.DisplayName -like '*Netskope*' }   # client
Get-ChildItem Cert:\LocalMachine\Root | Where-Object { $_.Subject -like '*Netskope*' }            # decryp
Resolve-DnsName www.example.com                                                                  # DNS w
```

Use `NSSWG-Validation.ps1` for a single pass/fail summary of the above.

H. When to escalate

If the device shows the client running, steering active, and the certificate trusted, but traffic is still not filtered as expected, collect the client logs and the policy/steering configuration and open a Netskope support case (Netskope tenant-side policy evaluation is outside device-level remediation).

7. Best Practices — Netskope Secure Web Gateway (SWG)

• **Pilot first, always.** The gateway is inline — a small `Web Steering – Pilot – Stage 1` scope catches wrong blocks and decryption-compatibility problems before they hit the fleet. This is the single most important practice.

• **Rehearse the rollback before you steer.** Know exactly how to remove a scope group from steering, disable `Web Filtering Policy – All Users – Production`, or uninstall the client — and time it.

• **Start decryption with a bypass list.** Never decrypt everything on day one. Bypass certificate-pinned apps and sensitive/regulated categories (health, banking) to avoid breakage and privacy issues.

• **Block the security categories, coach the borderline ones.** Block malware, phishing, and command-and-control outright; use coach (warn) for acceptable-use grey areas so users self-correct.

• **Deploy the certificate before (or with) the client.** HTTPS inspection throws certificate errors if `Decryption Certificate – Managed Devices – Production` is not trusted first.

• **Keep policy identity-based.** Scope `Web Filtering Policy – All Users – Production` to users/groups, not IP ranges, so it follows the user on and off network.

• **Order rules deliberately.** Put block rules before allow rules so a broad allow can't shadow a security block.

• **Watch coverage.** Run `NSSWG-Report.ps1` to confirm steered scope and per-category activity, and to catch devices that should be steered but are not.

• **Handle miscategorization quickly.** Maintain a process for category-correction requests and documented allow/block exceptions so user trust holds.

• **Communicate to users.** A short note about the block/coach pages and who to contact dramatically reduces tickets at rollout.

• **Document the artifacts.** Record the names and intent of the steering scope, categorization profile, filtering policy, and threat/content control for audits and repeatability.

8. Deliverables Summary — Netskope Secure Web Gateway (SWG)

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What an SWG is, steering, categorization, filtering, threat/content, on/off-network model, named artifacts, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Network)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps across Netskope, Entra, and Intune
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`NSSWG-Discovery.ps1`	Records SWG readiness: Entra steering groups, Intune client/cert deployment, and (optional) Netskope tenant readiness via REST. Read-only
`NSSWG-Validation.ps1`	On a device, confirms the Netskope client is running, steering is active, and the decryption certificate is trusted. Read-only
`NSSWG-Report.ps1`	Reports steering-scope coverage (Graph) and, optionally, per-category web activity from the Netskope REST API. Read-only
`NSSWG-Remediation.ps1`	Safely fixes mis-steered/non-compliant devices (re-sync, re-add to steering scope). Default `-WhatIf`
`NSSWG-Set-ScopeTargeting.ps1`	Creates/populates the `Web Steering – Pilot – Stage 1` and `Web Steering – Broad – Stage 2` groups (Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Every user's web traffic behind one cloud secure web gateway — categorized, filtered, and consistently enforced on or off network — proven on a pilot before it scales, with a documented rollback at every stage.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

NSSWG-Discovery.ps1

Purpose: Read-only readiness check. Reports whether the steering-scope Entra groups exist, whether the Intune client app and decryption-certificate profile are present, and (optionally) confirms Netskope tenant readiness via the Netskope REST API. Nothing is changed. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\NSSWG-Discovery.ps1 Microsoft side uses Microsoft Graph (read scopes below); sign in when prompted. Optional Netskope check: set environment variables NSSWG_NETSKOPE_BASEURL and NSSWG_NETSKOPE_TOKEN first (secrets are read from the environment, never hardcoded). EXPECTED RESULTS A readiness summary: steering groups present/absent, client app present, certificate profile present, and (if configured) a Netskope tenant-reachable indicator. .NOTES Workplace Marketplace — Network Solutions. Read-only. Netskope REST endpoints/paths are tenant- and version-specific and are NOT invented here; the Netskope call below is generic (caller supplies the base URL/path) and clearly marked. #> [CmdletBinding()] param([string]$PilotGroup = 'Web Steering - Pilot - Stage 1', [string]$BroadGroup = 'Web Steering - Broad - Stage 2', [string]$ClientAppName = 'SWG Client Deployment - Managed Devices - Production', [string]$CertProfileName = 'Decryption Certificate - Managed Devices - Production')

How to run: powershell -ExecutionPolicy Bypass -File .\NSSWG-Discovery.ps1 Microsoft side uses Microsoft Graph (read scopes below); sign in when prompted. Optional Netskope check: set environment variables NSSWG_NETSKOPE_BASEURL and NSSWG_NETSKOPE_TOKEN first (secrets are read from the environment, never hardcoded). EXPECTED RESULTS A readiness summary: steering groups present/absent, client app present, certificate profile present, and (if configured) a Netskope tenant-reachable indicator. .NOTES Workplace Marketplace — Network Solutions. Read-only. Netskope REST endpoints/paths are tenant- and version-specific and are NOT invented here; the Netskope call below is generic (caller supplies the base URL/path) and clearly marked. #> [CmdletBinding()] param([string]$PilotGroup = 'Web Steering - Pilot - Stage 1', [string]$BroadGroup = 'Web Steering - Broad - Stage 2', [string]$ClientAppName = 'SWG Client Deployment - Managed Devices - Production', [string]$CertProfileName = 'Decryption Certificate - Managed Devices - Production')

NSSWG-Remediation.ps1

Purpose: For a device that should be steered but is not, this re-adds it to a steering-scope Entra group and requests an Intune/MDM sync so the client app, certificate, and steering config re-apply. SAFE BY DEFAULT: runs in -WhatIf mode and only reports intended actions unless you pass -Apply. Inline proxy affects live traffic — review the plan before applying. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\NSSWG-Remediation.ps1 -DeviceId Apply: powershell -ExecutionPolicy Bypass -File .\NSSWG-Remediation.ps1 -DeviceId -Apply On-device sync only: run elevated with -LocalSync on the affected device. EXPECTED RESULTS With no -Apply: a printed plan (what would change). With -Apply: the device is added to the target steering group and a sync is requested. Re-run NSSWG-Validation.ps1 afterward. .NOTES Workplace Marketplace — Network Solutions. Mutating actions are gated behind -Apply. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([string]$DeviceId, [string]$TargetGroup = 'Web Steering - Broad - Stage 2', [switch]$Apply, [switch]$LocalSync)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\NSSWG-Remediation.ps1 -DeviceId <entraDeviceId> Apply: powershell -ExecutionPolicy Bypass -File .\NSSWG-Remediation.ps1 -DeviceId <entraDeviceId> -Apply On-device sync only: run elevated with -LocalSync on the affected device. EXPECTED RESULTS With no -Apply: a printed plan (what would change). With -Apply: the device is added to the target steering group and a sync is requested. Re-run NSSWG-Validation.ps1 afterward. .NOTES Workplace Marketplace — Network Solutions. Mutating actions are gated behind -Apply. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([string]$DeviceId, [string]$TargetGroup = 'Web Steering - Broad - Stage 2', [switch]$Apply, [switch]$LocalSync)

NSSWG-Report.ps1

Purpose: Reports SWG coverage: the steering-scope Entra groups and their member counts (Microsoft Graph), plus the client-app assignment targets. Optionally pulls per-category web activity from the Netskope REST API to confirm traffic is flowing through the gateway. Read-only. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\NSSWG-Report.ps1 -CsvPath C:\Temp\nsswg.csv Microsoft side uses Microsoft Graph (read scopes below); sign in when prompted. Optional Netskope activity: set NSSWG_NETSKOPE_BASEURL and NSSWG_NETSKOPE_TOKEN, and pass -NetskopeActivityPath ''. EXPECTED RESULTS Steering-group coverage (members per group), client-app assignment targets, an optional CSV, and (if configured) a per-category web-activity summary from the Netskope tenant. .NOTES Workplace Marketplace — Network Solutions. Read-only. The Netskope REST path is tenant/version specific and is NOT invented here — supply it yourself via -NetskopeActivityPath per Netskope's API documentation. Secrets come from the environment. #> [CmdletBinding()] param([string]$PilotGroup = 'Web Steering - Pilot - Stage 1', [string]$BroadGroup = 'Web Steering - Broad - Stage 2', [string]$ClientAppName = 'SWG Client Deployment - Managed Devices - Production', [string]$CsvPath, [string]$NetskopeActivityPath)

How to run: powershell -ExecutionPolicy Bypass -File .\NSSWG-Report.ps1 -CsvPath C:\Temp\nsswg.csv Microsoft side uses Microsoft Graph (read scopes below); sign in when prompted. Optional Netskope activity: set NSSWG_NETSKOPE_BASEURL and NSSWG_NETSKOPE_TOKEN, and pass -NetskopeActivityPath '<exact read-only REST path for your tenant/version>'. EXPECTED RESULTS Steering-group coverage (members per group), client-app assignment targets, an optional CSV, and (if configured) a per-category web-activity summary from the Netskope tenant. .NOTES Workplace Marketplace — Network Solutions. Read-only. The Netskope REST path is tenant/version specific and is NOT invented here — supply it yourself via -NetskopeActivityPath per Netskope's API documentation. Secrets come from the environment. #> [CmdletBinding()] param([string]$PilotGroup = 'Web Steering - Pilot - Stage 1', [string]$BroadGroup = 'Web Steering - Broad - Stage 2', [string]$ClientAppName = 'SWG Client Deployment - Managed Devices - Production', [string]$CsvPath, [string]$NetskopeActivityPath)

NSSWG-Set-ScopeTargeting.ps1

Purpose: Creates the two steering-scope security groups used to stage the rollout — 'Web Steering - Pilot - Stage 1' and 'Web Steering - Broad - Stage 2' — and optionally adds members from a CSV of object ids. These groups scope which devices/users are steered into the gateway and are referenced by the Intune client/cert assignments and the Netskope steering configuration. HOW TO RUN Create groups only: powershell -ExecutionPolicy Bypass -File .\NSSWG-Set-ScopeTargeting.ps1 Create and add members (CSV with a header 'ObjectId'): powershell -ExecutionPolicy Bypass -File .\NSSWG-Set-ScopeTargeting.ps1 -PilotMembersCsv .\pilot.csv -BroadMembersCsv .\broad.csv EXPECTED RESULTS Both groups exist (created if missing) and any supplied members are added. The

names are emitted so you can reference them verbatim in Intune and Netskope. .NOTES Workplace Marketplace — Network Solutions. Creates groups and adds members (mutating) — uses -WhatIf via ShouldProcess; pass -Confirm:$false to skip prompts in automation. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'Medium')] param([string]$PilotGroup = 'Web Steering - Pilot - Stage 1', [string]$BroadGroup = 'Web Steering - Broad - Stage 2', [string]$PilotMembersCsv, [string]$BroadMembersCsv)

How to run: Create groups only: powershell -ExecutionPolicy Bypass -File .\NSSWG-Set-ScopeTargeting.ps1 Create and add members (CSV with a header 'ObjectId'): powershell -ExecutionPolicy Bypass -File .\NSSWG-Set-ScopeTargeting.ps1 -PilotMembersCsv .\pilot.csv -BroadMembersCsv .\broad.csv EXPECTED RESULTS Both groups exist (created if missing) and any supplied members are added. The names are emitted so you can reference them verbatim in Intune and Netskope. .NOTES Workplace Marketplace — Network Solutions. Creates groups and adds members (mutating) — uses -WhatIf via ShouldProcess; pass -Confirm:$false to skip prompts in automation. #> [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'Medium')] param([string]$PilotGroup = 'Web Steering - Pilot - Stage 1', [string]$BroadGroup = 'Web Steering - Broad - Stage 2', [string]$PilotMembersCsv, [string]$BroadMembersCsv)

NSSWG-Validation.ps1

Purpose: Verifies the Netskope client/service is running, traffic steering is active, and the decryption (root) certificate is trusted in the machine store — the three device-side prerequisites for the gateway to filter this device's web traffic. HOW TO RUN Elevated: powershell -ExecutionPolicy Bypass -File .\NSSWG-Validation.ps1 EXPECTED RESULTS "STEERING ACTIVE" when the client is running and the decryption certificate is trusted; otherwise a list of what is missing and the next step. .NOTES Workplace Marketplace — Network Solutions. Read-only. Service/process names below are matched generically; confirm the exact client identity for your Netskope client build in Netskope's documentation if the match is empty. #> [CmdletBinding()] param([string]$CertSubjectMatch = 'Netskope') Write-Host "=== Netskope SWG Validation — $($env:COMPUTERNAME) ===" -ForegroundColor Cyan

How to run: Elevated: powershell -ExecutionPolicy Bypass -File .\NSSWG-Validation.ps1 EXPECTED RESULTS "STEERING ACTIVE" when the client is running and the decryption certificate is trusted; otherwise a list of what is missing and the next step. .NOTES Workplace Marketplace — Network Solutions. Read-only. Service/process names below are matched generically; confirm the exact client identity for your Netskope client build in Netskope's documentation if the match is empty. #> [CmdletBinding()] param([string]$CertSubjectMatch = 'Netskope') Write-Host "=== Netskope SWG Validation — $($env:COMPUTERNAME) ===" -ForegroundColor Cyan